NATIONAL MATHEMATICS CONTEST
A-LEVEL PAPER 2 QUESTIONS AND SOLUTIONS.

SATURDAY 28TH JUNE, 2025 9:00 AM - 12:15 PM

SECTION A: 5 marks each

1. Evaluate the following expression: $\dfrac{1^4+2025^4+2026^4}{1^2+2025^2+2026^2}$.

Solution : *Consider the function* $f : \mathbb{R} \to \mathbb{R},\ x \mapsto \dfrac{1+x^4+(x+1)^4}{1^2+x^2+(x+1)^2}$. *Upon simplification,* $f(x) = \dfrac{1^4+x^4+(x+1)^4}{1^2+x^2+(x+1)^2} = \dfrac{1+2x+3x^2+2x^3+x^4}{1+x+x^2} = 1+x+x^2$, *by long division. Setting* $x = 2025$ *gives* $\dfrac{1^4+2025^2+2026^4}{1+2025^2+2026^2} = f(2025) = 1+2025+2025^2 = 4102651$. ■

2. Suppose that a, b, c, d are positive real numbers satisfying $(a+c)(b+d) = ac+bd$. Find the smallest possible value of $R = \dfrac{a}{b}+\dfrac{b}{c}+\dfrac{c}{d}+\dfrac{d}{a}$.

Solution : **Answer***: The smallest possible value of* R *is* 8.

- *To show that* $R \geq 8$, *we apply the* **AM-GM inequality** *twice as follows:*

$$R = \left(\frac{a}{b}+\frac{c}{d}\right)+\left(\frac{b}{c}+\frac{d}{a}\right) \geq 2\sqrt{\frac{ac}{bd}}+2\sqrt{\frac{bd}{ac}} = \frac{2(ac+bd)}{\sqrt{abcd}} = \frac{2(a+c)(b+d)}{\sqrt{abcd}} \geq \frac{2\cdot 2\sqrt{ac}\cdot 2\sqrt{bd}}{\sqrt{abcd}} = 8.$$

- *The above inequalities turn into equalities when* $a = c$ *and* $b = d$. *Then the condition* $(a+c)(b+d) = ac+bd$ *can be rewritten as* $4ab = a^2+b^2$. *So, it is satisfied when* $a/b = 2 \pm \sqrt{3}$. *Hence,* R *attains the value* 8, *when* $a = c = 1$ *and* $b = d = 2+\sqrt{3}$, *as required.*

■

3. Let x and y be positive integers such that $\dfrac{1}{x}+\dfrac{1}{y} = \dfrac{1}{45}$. What is the maximum possible value of y?

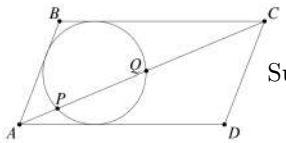
Solution 1 : *Rearranging* $\dfrac{1}{x}+\dfrac{1}{y} = \dfrac{1}{45}$ *gives* $\dfrac{1}{y} = \dfrac{1}{45}-\dfrac{1}{x} = \dfrac{x-45}{45x}$. *That is,*

$$y = \frac{45x}{x-45} = \frac{45x-2025+2025}{x-45} = 45+\frac{2025}{x-45}.$$

In order to maximize y, *we require* $x-20$ *to be as small as possible. Since* x *is an integer, then* $x = 46$. *Thus,* $y = 45+\dfrac{2025}{46-45} = 45+2025 = 2070$. *Therefore,* $\Rightarrow y_{\max} = 2070$. ■

Solution 2 : *Rearranging* $\dfrac{1}{x}+\dfrac{1}{y} = \dfrac{1}{45}$ *gives* $\dfrac{x+y}{xy} = \dfrac{1}{45}$, *which implies that* $xy = 45x+45y$ *or* $xy-45x-45y = 0$. *This implies that* $(x-45)(y-45) = 2025 = 1\cdot 2025$ *is a possible factorization. Thus,* $(x-45) = 1 \Rightarrow x_{\min} = 46$ *and* $(y-45) = 2025 \Rightarrow y_{\max} = 2070$. ■

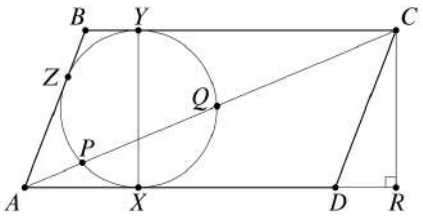
4. Let $ABCD$ be a parallelogram with $\angle BAD < 90^o$. A circle tangent to sides $\overline{DA}$, $\overline{AB}$, and $\overline{BC}$ intersects diagonal $\overline{AC}$ at points P and Q with $AP < AQ$ as shown in the figure on the right.



Suppose that $AP = 3$, $PQ = 9$, and $QC = 16$. Then the area of $ABCD$ can be expressed in the form $m\sqrt{n}$, where m and n are positive integers, and n is not divisible by the square of any prime. Find the value of $m+n$.

Solution : **Answer***: The required value is* $m+n = 150$.

- *Let* X, Y, *and* Z *denote the points where the circle is tangent to* $\overline{AD}$, $\overline{BC}$, *and* $\overline{AB}$, *respectively. Let* R *be the foot of the perpendicular from* C *to* $\overline{AD}$.



- *By* **Power of a Point**, *we have* $AX = \sqrt{AP\cdot AQ} = 6$ *and* $CY = \sqrt{CQ\cdot CP} = 20$. *Hence,* $AR = AX+CY = 26$.
- *By the* **Pythagorus Theorem**, *we have* $XY = CR = \sqrt{AC^2-AR^2} = \sqrt{28^2-26^2} = \sqrt{108} = 6\sqrt{3}$.
- *Let* $BY = BZ = t$. *Then* $AB = 6+t$, *and so again by the* **Pythagorean Theorem**

$$6\sqrt{3} = XY = \sqrt{(6+t)^2-(6-t)^2} = 2\sqrt{6t}.$$

- *Thus, $t = \frac{9}{2}$, so $BC = \frac{49}{2}$, and the area of $ABCD$ is $\frac{49}{2} \cdot 6\sqrt{3} = 147\sqrt{3}$. So, the required sum is $m + n = 147 + 3 = 150$.*

■

5. What is the largest positive integer n such that $n^3 + 2025$ is divisible by $n + 45$?

Solution : **Answer**: *The largest possible value of n is* $\boxed{n = 89055}$.

- *Either do* **long division** *or: Let*

$$n^3 + 2025 = (n + 45)(n^2 + an + b) + c = n^3 + n^2(45 + a) + n(b + 45a) + 45b + c.$$

Now, equating the coefficients yields the following system of equations:

$$45 + a = 0, \quad b + 45a = 0, \quad 45b + c = 2025.$$

Solving the system yields: $a = -45$, $b = 2025$, and $c = -89100$. Thus,

$$n^3 + 2025 = (n + 45)(n^2 - 45n + 2025) - 89100.$$

- *Therefore, by the* **Euclidean Algorithm**, *we get:*

$$n + 45 = \gcd(n^3 + 2025, n + 45) = \gcd(-89100, n + 45) = \gcd(89100, n + 45),$$
$$\Rightarrow n + 45 = 89100 \quad \Rightarrow n = 89055.$$

The maximum value for n is hence $\boxed{n = 89055}$.

■

6. A head of a school in Kampala says that the percentage of women teachers in the school is somewhere strictly between 60 and 65 percent. What is the smallest number of employees that could be in the school?

Solution : *Let K be the number of female(women) employees among the n- employees, this comes down to considering what possible percentages are there with fraction $\frac{K}{n}$ for a fixed n say if $n = 3$, possible fractions are $0\%, 33\%, 67\%$, and 100% none of them falls in the required range trying this way for $n = 4, 5, 6, 7, 8, \ldots$ we have that for $n = 8$ as the smallest value for n such that the required $\%$ is between 60 and 65 that is when $K = 6$ and $n = 8$ thus $\frac{6}{8} = 62.5\% \Rightarrow$ the smallest number of employees is 8.* ■

7. Let $n \geq 40$ be an integer. On a chalkboard we write in a line n numbers each being equal to $+1$ or -1 such that:

(i) The sum of every 40 successive numbers is equal to 0.

(ii) The sum of every 42 consecutive numbers is not equal to 0.

We denote by $\sum_n$ the sum of the n numbers on the chalkboard. Find the greatest possible value of $\sum_n$.

Solution :

- *Let $a_1, a_2, \ldots, a_n$ be the numbers written in line on the chalkboard. Since $a_1 + a_2 + \cdots + a_{40} = 0$, it follows that the half of the numbers $a_1 + a_2 + \cdots + a_{40} = 0$ are equal to $+1$ and the other half are equal to -1.*
- *Since $a_1 + a_2 + \cdots + a_{40} + a_{42} + a_{42} \neq 0$, it follows that $a_{41} = a_{42} = a \in \{+1, -1\}$. Similarly, $a_1 + a_2 + \cdots + a_{43} \neq 0$ and hence, $a_{42} = a_{43} = a \in \{+1, -1\}$. Proceeding in the same way as before, we conclude that all the numbers after a_{41} must be equal to a.*
- *Therefore, if $n > 60$, the sum $a_{22} + a_{23} + \cdots + a_{61}$ has $a_{41} = a_{42} = \cdots = a_{61} = a$, and hence $+1$ and -1 cannot have the same cardinality in the set. This means that $a_{22} + a_{23} + \cdots + a_{61} \neq 0$, and this contradicts condition (i) of the problem statement.*
- *Hence, the greatest value of n is 60.*
- *The maximum value of $\sum_n$ is obtained when we have as many as possible numbers equal to $+1$. We remind ourselves that between the first 40 numbers we must have 20 numbers equal to -1. Hence the numbers equal to $+1$ is total can be at most 40. Therefore, $\sum_n \leq 40 - 20 = 20$.*
- *In fact, we observe that $\max \sum_n = 20$ and it happens for $a_1 = a_2 = \cdots = a_{20} = +1$, $a_{21} = a_{22} = \cdots = a_{40} = -1$, and $a_{41} = a_{42} = \cdots = a_{60} = +1$.*

■

8. Let a, b, c be arbitrary integers. Define the function $f(x) = ax^2 + bx + c$. Prove that the expression

$$f(80) + f(79) + \cdots + f(41) - f(40) - f(39) - \cdots - f(1),$$

is divisible by 80.

Solution :

- *Let us denote and rewrite the expression in question as follows: $S = f(80) + f(79) + \cdots + f(41) - f(40) - f(39) - \cdots - f(1)$ $= a(80^2 + 79^2 + \cdots + 41^2 - 40^2 - 39^2 - \cdots - 1^2)$ $+ b(80 + 79 + \cdots + 41 - 40 - 39 - \cdots - 1) = S_1 + S_2$. Note that the c cancels out during the substitution process!*

- *It remains to show that S_1 and S_2 are both divisible by 80: $S_1 = a((80^2-40^2)+(79^2-1^2)+(78^2-2^2)+\cdots+(41^2-39^2))$*
  $= a(120\cdot 40+80\cdot 78+80\cdot 76+\cdots+80\cdot 2)$
  $= a(80\cdot 60+80\cdot 78+80\cdot 76+\cdots+80\cdot 2)$
  $= 80a(60+78+76+\cdots 2) \quad \Rightarrow \quad 80 \mid S_1.$
- *Similarly,*

$$\begin{aligned} S_2 =& b(80+79+\cdots+41-40-39-\cdots-1) \\ &= b(80-40+79-39+\cdots+41-1) = b(40+40+\cdots- \\ &= 40\cdot 40b = 80\cdot 20b \quad \Rightarrow \quad 80 \mid S_2. \end{aligned}$$

■

9. Erina has twelve blocks, two each of red (R), blue (B), yellow (Y), green (G), orange (O), and purple (P). Call an arrangement of blocks even if there is an even number of blocks between each pair of blocks of the same color. For example, the arrangement

$$RBBYGGYROPPO$$

is even. Erina arranges her blocks in a row in random order. The probability that her arrangement is even is $\frac{m}{n}$, where m and n are relatively prime positive integers. Find the value of $m+n$.

Solution : **Answer**: $m+n=247$.

- *More generally, let there be k colors, numbered $1,2,\ldots,k$, with two blocks of each color. Note that an arrangement of blocks is even if and only if every pair of blocks of the same color are in positions whose position numbers differ by an odd number, which means that one position number is even and the other is odd.*
- *Thus Erina forms an even arrangement if and only if there are two permutations of the k colors $a_1,a_2,\ldots,a_k$ and $b_1,b_2,\ldots,b_k$ such that Erina arranges the blocks in the order $a_1,b_1,a_2,b_2,\ldots,a_k,b_k$.*
- *It follows that there is a total of $(k!)^2$ even arrangements. The number of ways to arrange k pairs of blocks is $\frac{(2k)!}{2^k}$.*
- *For $k=6$, the probability that a random arrangement is even equals*

$$\frac{(6!)^2\cdot 2^6}{12!} = \frac{2\cdot 4\cdot 6}{7\cdot 9\cdot 11} = \frac{16}{231}.$$

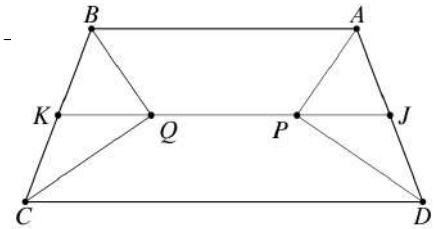
- *Thus, the required sum is $m+n = 16+231 = 247$.*

■

10. In an isosceles trapezoid $ABCD$, parallel bases $\overline{AB}$ and $\overline{CD}$ have lengths $500m$ and $650m$, respectively, and the lines $\overline{AD}$ and $\overline{BC}$ both have lengths $333m$. The angle bisectors of $\angle A$ and $\angle D$ meet at P , and the angle bisectors of $\angle B$ and $\angle C$ meet at Q. Find the length of $\overline{PQ}$.

Solution : **Answer**: *The length of $\overline{PQ}$ is $242m$.*

Consider the following diagram.



- *Let K lie on $\overline{BC}$ so that $\overline{QK} \parallel \overline{AB}$. Since $\angle KBQ = \angle QBA = \angle BQK$, it follows that $\triangle KBQ$ is isosceles with $BK = QK$. Similary, $\triangle KCQ$ is isosceles with $CK = QK$. Hence, $BK = QK = CK$.*
- *Analogously, let point J lie on $\overline{AD}$ so that $\overline{PJ} \parallel \overline{AB}$. Then J is the midpoint of $\overline{AD}$, and $AJ = PJ = DJ$. Thus, points P and Q lie of $\overline{KJ}$, which is the midline of the trapezoid with $\overline{KJ} \parallel \overline{AB}$, and*

$$KJ = \frac{AB+CD}{2} = \frac{500+650}{2} = 575m.$$

- *Therefore, $PQ = KJ - KQ - PJ = KJ - 2\cdot PJ = KJ - AD = 575 - 333 = 242m$.*

Remark: *An alternative solution is possible using similarity of triangles.* ■

SECTION B: 10 marks each

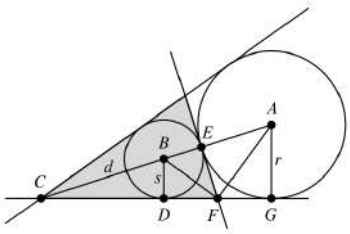
11. A man has two identical jars. The first is full of water, while the second is empty. He pours half the water from the first jar into the second. Then on the second transfer, he pours one third of the water that is in the second jar back into the first. He repeats this, alternating cups, pouring $\dfrac{1}{(i+1)}$ of the water that is in one jar back into the other on the ith transfer. What fraction of the water is in the first jar just before the 2025th transfer?

Solution : *Working out the first few times it shows that after an odd numbered transfer it will be half and half however, just before the $(2k+1)th$ transfer it will be $\frac{k+1}{2k+1}$ in the first jar and $\frac{k}{2k+1}$ in the second jar. check this for $k=1,2,3,\ldots$, now since 2025 is odd so just before 2025 the first jar has $\frac{2025+1}{2(2025)+1} = \frac{2026}{4051}$.*
■

12. A circle with radius 6 *cm* is externally tangent to a circle with radius 24 *cm*. Find the area of the triangular region bounded by the three common tangent lines of these two circles.

Solution : **Answer***: The required area is* 192 cm^2.

- *More generally, let the larger circle have radius r and center A and the smaller circle have radius s and center B. Let the two circles be tangent at E, let the common external tangents intersect at C , let one of those tangents be tangent to the larger circle at G and to the smaller circle at D, let that tangent intersect the common internal tangent at F , and let $d = BC$, as shown. in the figure below.*



- *Since $\triangle CBD$ and $\triangle CAG$ are similar, we have:*

$$\frac{d}{s} = \frac{d+s+r}{r},$$

from which we get that $d = s\left(\frac{r+s}{r-s}\right)$.

- *Since $\angle DFE$ and $\angle EFG$ are supplementary, and $\overline{BF}$ and $\overline{AF}$ bisect these angles, $\triangle AFB$ is a right-angled triangle, so $FE = \sqrt{rs}$.*
- *The required area is then given by*

$$(d+s)\sqrt{rs} = \frac{2rs\sqrt{rs}}{r-s}.$$

- *Substituting for $r = 24$ and $s = 6$ gives the required area as:*

$$\frac{2 \cdot 24 \cdot 6\sqrt{24 \cdot 6}}{24-6} = 192 \ cm^2.$$

■

13. Find the last three digits of the number

$$2025^1 + 2025^2 + 2025^3 + \cdots + 2025^{2026}.$$

Solution : *This is done by reducing the expression modulo* 1000*, it remains to find the last three digits of the somewhat-less-daunting expression. That is,*

$$2025^1 + 2025^2 + 2025^3 + \cdots + 2025^{2026} \equiv 25^1 + 25^2 + 25^3 + \cdots + 25^{2026} \ (mod\ 1000).$$

Observe that the last three digits in each of the individual terms in the sum are: $25^1 = 25 \equiv 025$, $25^2 = 625 \equiv 625$, $25^3 = 15625 \equiv 625, \ldots, 25^{2026} \equiv 625 (mod 1000)$. *Apart from* 25^1 *which ends in* 025*, the rest of the terms have* 625 *as the last three digits in their decimal expansion. Now, summing up the last three digits in the individual terms together, we get that:* $25^1+25^2+25^3+\cdots+25^{2026} \equiv 025+625\cdot(2026-1) \equiv 025+625 \equiv 650 \ (mod\ 1000)$. *Therefore,* $2025^1 + 2025^2 + 2025^3 + \cdots + 2025^{2026} \equiv \boxed{650} \ (mod\ 1000)$. ■

14. Let $x_1, x_2, x_3, \ldots, x_{2025}$ be distinct positive real numbers such that

$$a_n = \sqrt{(x_1 + x_2 + x_3 + \cdots + x_n)\left(\frac{1}{x_1} + \frac{1}{x_2} + \frac{1}{x_3} + \cdots + \frac{1}{x_n}\right)}$$

is an integer for every $n = 1, 2, 3, \ldots, 2025$. Prove that $a_{2025} \geq 3037$.

Solution : *We start by making some basic observations:*

- *First note that the sequence $a_1, a_2, a_3, \ldots, a_{2025}$ is increasing and thus, since all elements are integers, $a_{n+1} - a_n \geq 1$.*
- *We also observe that $a_1 = 1$ and*

$$a_2 = \sqrt{(x_1 + x_2)\left(\frac{1}{x_1} + \frac{1}{x_2}\right)} > 2,$$

by the **Cauchy-Schwarz inequality** *and using $x_1 \neq x_2$. So, $a_3 \geq 3$.*

Now, we proceed to the main part of the argument.

- *We observe that* 3037 *is about three halves of* 2025*. Motivated by this observation, we prove the following claim:*
- **Claim***: If $a_{n+1} - a_n = 1$, then $a_{n+2} - a_{n+1} \geq 2$. In otherwords, the sequence has to increase by at least* 2 *at at least half of the times. Assuming that the claim is true, since $a_1 = 1$, we we would be done since:* $a_{2025} = (a_{2025} - a_{2024}) + (a_{2024} - a_{2023}) + \cdots + (a_2 - a_1) + a_1 \geq (2+1) \cdot \frac{2024}{2} + 1 = 3 \cdot 1012 + 1 = 3037$. *Clearly, if the claim is true, then $a_{2025} \geq$* 3037*.*

Proof of the Claim: *We start by observing that*

$$\begin{aligned} a_{n+1}^2 &= (x_1 + x_2 + \cdots + x_{n+1})\left(\frac{1}{x_1} + \frac{1}{x_2} + \cdots + \frac{1}{x_{n+1}}\right) \\ &= (x_1 + \cdots + x_n)\left(\frac{1}{x_1} + \cdots + \frac{1}{x_n}\right) + 1 + \frac{1}{x_{n+1}}(x_1 + \cdots + x_n) + x_{n+1}\left(\frac{1}{x_1} + \cdots + \frac{1}{x_n}\right) \\ &\geq a_n^2 + 1 + 2\sqrt{\frac{1}{x_{n+1}}(x_1 + \cdots + x_n) \cdot x_{n+1}\left(\frac{1}{x_1} + \cdots + \frac{1}{x_n}\right)} = a_n^2 + 1 + 2a_n = (a_n + 1)^2, \end{aligned}$$

where we have used the **AM-GM inequality** *to obtain the inequality.*
In particular, if $a_{n+1} = a_n + 1$, *then*

$$\frac{1}{x_{n+1}}(x_1 + \cdots + x_n) = x_{n+1}\left(\frac{1}{x_1} + \cdots + \frac{1}{x_n}\right) \tag{1.1}$$

We assume for contradiction that both $a_{n+1} = a_n + 1$ *and* $a_{n+2} = a_{n+1} + 1$ *hold. It is very easy to see that using* (1.1) *for both cases, gives:* $x_{n+2} = x_{n+1}$, *which is an obvious contradiction.* ■

15. Eight children, who did not know each other, came to the dance class. To introduce them to each other, the teacher decided to choose four children each minute and let them dance in a circle. During this minute, each of the children in a circle will get to know the neighbour on the left and the right. What is the minimum number of minutes the teacher needs to make all the eight children know each other.

Solution : **Answer**: 8 *minutes.*

- *Consider any child from this group. They need to get to know seven other children, and participating once in a dancing circle they meet a maximum of two new acquaintances.*
- *Thus, each child sould participate in at least four dancing circles. In each dancing circle, there are not more than four children participating. Then, there were at least* $\frac{8 \cdot 4}{2} = 8$ *circles.*
- *Since the teacher needs one minute for each circle, then the minimum number of minutes needed by the teacher to make all the children get to know each other is* $8 \cdot 1 = 8$ *minutes.*

■

END